Team Name/Number: _____

Science Olympiad Thermodynamics test
Kenston Invitational 1/13/18

Each question is worth 1 point unless indicated otherwise. 25 possible points.

1. The lowest possible temperature is ____.
   a. 0 °C b. 0 ºK c. –273 °F d. –273 °K

2. The hottest possible temperature is
   a. infinity b. 1,000,000 K c. 9,999,999,999 K d. 1,000 K

3. Fast food restaurants keep food hot with infrared lamps. The heat is transferred by ____.
   a. condensation b. conduction c. convection d. Radiation

4. When a pot of water is put on a stove, the water at the top gets hot primarily by ____.
   a. radiation b. conduction c. convection d. condensation

5. A small candy bar has 52 calories. If all of its energy were converted to heat, by how much would it raise the temperature of 1 L of water?
   a. 1° C b. 104° C c. 52° C d. 0° C

6. An ideal gas is compressed isobarically to one-third of its initial volume. The resulting pressure will be
   A) 3 times as large as the initial value. B) equal to the initial value.
   C) more than 3 times as large as the initial value. D) 9 times the initial value.

7. If, under steady state heat flow, you double the thickness of a wall built from solid uniform material, the rate of heat loss for a given temperature difference across the thickness will?
   A) become ½ its original value. B) also double.
   C) become ¼ its original value. D) become 3/2 of its original value.

8. Around 1850 _____ was/were the first to state the First Law of Thermodynamics.
   a. Rudolf Clausius / William Thomson (Kelvin) b. William Gibbs & Enrico Fermi
   c. Daniel Bernoulli & Ludwig Boltzmann d. Martin Van Buren

9. Who discovered the mechanical equivalent of heat?
   a. Robert Boyle c. Antoine Lavoisier
   b. James Joule d. James Watt

10. In what year was the concept of Absolute Zero first postulated?
    a. 1949 c. 1492
    b. 1835 d. 1702

11. This was a hypothetical substance thought to exist in all combustible objects and released when burned -- this is now an obsolete idea.
    a. Phlogiston b. Newnonium
    c. Splunkston d. Obsoletonium

12. Known as the father of thermodynamics, this person gave the first successful theory of the maximum efficiency of heat engines
    a. Robert Boyle c. William Thomson (1st Baron Kelvin)
    b. Nicolas Léonard Sadi Carnot d. Josiah Willard Gibbs

13. (5 pts.) Consider the laws of thermodynamics. For each of the following statements, which law of thermodynamics (0th, 1st, 2nd, or 3rd) is best illustrated?

    a) A heat engine works by transferring heat between 2 reservoirs _____
    b) The energy stored in a Candle goes mostly to heat and light _____
    c) A thermometer is placed in a cup of hot water to measure its temperature. _____
    d) Coffee is placed in a coffee mug. The outside of the mug heats up. _____
    e) An oscillating pendulum eventually stops and comes to rest. _____

14. A comfortable summer day is 77°F. What is this on the Celsius scale?

15. (2 pts.) There is one special temperature where the temperature in Fahrenheit and Celsius are the same number. What is that temperature?

16. The Rankine (R) temperature scale is defined analogously to the Kelvin scale, but instead of each degree increment being equal to 1 degree celsius it is equal to 1 degree Fahrenheit. For example, the lowest possible temperature (absolute zero) is 0 °R, while the freezing point of water occurs at 491.67 °R. So here's the question: at what Rankine temperature does water boil?

17. **Completion: _____** is the amount of energy absorbed or released during a phase-change process.

18. **Completion:** The amount of heat required to raise the temperature of 1 kg of a substance by 1°C is its _____.

19. (2pts.) Suppose that there is a sample of ideal gas occupying 5.00 L at a pressure of 2.00 atm. The temperature is determined to be 150 ºF. Suppose now that the gas is made to contract until the volume is 3.50 L, while keeping a constant pressure of 2.00 atm. Calculate the work done **on the gas** in this process. (1 atm = 10^5 Pa $= 10^5\ N/m^2$)